Quantitative Reflection and Trace Certificates for Weak Mixing of Irrational Triangular Billiards

Dr. Muralidhara Subbarao
Technical Report Assisted by AI Models
www.integralresearch.net
Stony Brook University
Technical Report TR-Ram-OpenAI-722-M287

October 8, 2026

Abstract

The source manuscript claims weak mixing of the full Liouville billiard flow in every triangle with an irrational angle. Its decisive analytic argument upgrades bounded measurable eigenfunctions to transverse rigidity through localized energy estimates and two ordered limits. We retain that argument as the upstream global input and develop a quantitative version of its final reflection obstruction. An explicit angular-tail certificate, a global lower bound for the third-side phase defect, and triangle-specific Sobolev trace estimates yield a computable exclusion inequality for a normalized field at nonzero frequency. The phase bound covers an entire unbounded frequency range when the other data are uniform. A rational triangle with an irrational angle supplies exact worked constants. We also prove a small-divisor obstruction to tail-free certification, state geometry and measurement error budgets, and connect finite Ram moments to physical Gram and trace norms. These are conditional stability certificates, not a rate for the source's singular-limit argument or a finite verification of all eigenfrequencies. Contribution assessment: C3/5, the report establishes a source-specific quantitative reflection-to-trace exclusion theorem with explicit global phase margins and an exact geometric example, while the global billiard rigidity argument remains an independent obligation.

1 Problem, measures, and scope

Let $Q \subset \mathbb{R}^2$ be a nondegenerate open triangle of area A and diameter D. Write $v(\theta) = (\cos\theta, \sin\theta)$, $v^{\perp}(\theta) = (-\sin\theta, \cos\theta)$, and

$$\mathrm{d}\sigma(\theta) = \frac{\mathrm{d}\theta}{2\pi}, \qquad \mathrm{d}\mu(x,\theta) = \frac{\mathrm{d}x}{A}\,\mathrm{d}\sigma(\theta), \qquad X = v\cdot\nabla_x, \quad Y = v^{\perp}\cdot\nabla_x.$$

All unqualified circle norms below use σ, and all interior norms use μ. Edge norms use normalized arclength and σ. These are probability normalizations, not unnormalized boundary integrals. Complex Hilbert inner products are linear in their first argument. Frequencies λ are real.

Remove the Liouville-null set of billiard trajectories that encounter a vertex in either time direction. The source concerns the resulting *full* unit-speed billiard flow, not a directional flow on a rational unfolding. A flow eigenfunction satisfies

$$F \circ \Phi_t = \mathrm{e}^{i\lambda t} F \quad \text{in } L^2(\mu) \quad \text{for every } t \in \mathbb{R}. \tag{1}$$

Weak mixing requires that all such eigenfunctions be constant (and have $\lambda = 0$ unless zero). Ergodicity alone addresses invariant functions, not the nonzero frequencies. Testing finitely many λ does not settle this assertion. Statements about a dense G_δ set of polygons, or about almost every direction in a rational polygon, have different quantifiers.

The cached October 5 source [1] was read in full, including all five scientific modules. We give its dependency map in Section 2. The proofs of the new estimates in Sections 3–7 are self-contained. No quantitative transverse estimate for arbitrary approximate eigenfunctions is imported or silently inferred from the source. In particular, the transverse term YF in our certificate must actually be bounded. A small value of $(X - i\lambda)F$ alone is insufficient.

2 Source proof architecture and exact gates

The following account distinguishes reading the supplied proof from an independent certification of the global dynamical theorem or its literature. No mathematical correction to the scoped source chain was identified here; several easily lost restrictions are essential to its validity.

2.1 Doubling, operator domains, and modes

The source doubles two oppositely oriented copies of Q along open sides. Removing the three cone points gives an oriented flat surface M; its completion has cone angles twice the corresponding triangle angles. The regular geodesic flow Ψ_t exists almost everywhere for all times and folds to the billiard flow. The justification includes the nullity of vertex hits and exclusion of an accumulation of seam crossings away from a cone. The induced unitary group is strongly continuous. These statements are not consequences of a finite angular matrix.

In a flat chart put

$$\partial = \tfrac{1}{2}(\partial_x - i\partial_y), \quad \bar\partial = \tfrac{1}{2}(\partial_x + i\partial_y), \quad X = \mathrm{e}^{i\theta}\partial + \mathrm{e}^{-i\theta}\bar\partial, \quad Y = i\mathrm{e}^{i\theta}\partial - i\mathrm{e}^{-i\theta}\bar\partial.$$

Here $[X, Y] = 0$ on the regular surface; initially every distributional test has compact support away from the cone points. A mode $u_j \mathrm{e}^{ij\theta}$ transforms as a flat unitary line bundle under changes of chart, not as an arbitrary globally scalar coefficient. The mode identities are

$$(Xu)_j = \partial u_{j-1} + \bar\partial u_{j+1}, \qquad (Yu)_j = i\partial u_{j-1} - i\bar\partial u_{j+1}. \tag{2}$$

The horizontal identity $\|\nabla u\|^2 = \|Xu\|^2 + \|Yu\|^2 = \sum_j \|\nabla u_j\|^2$ is used only once the derivatives exist in L^2. It is not an initial regularity assumption on an eigenfunction.

2.2 Localized energy and the two limits

For $\mathcal{A} = X - i\lambda$, a smooth, spatially compactly supported regular field w with $\mathcal{A}w$ supported over E obeys the source estimate

$$\tfrac{1}{2}D_m + D_{m+1} \le \|\mathcal{A}w\| \, \|1_E \nabla w\| + \tfrac{\lambda^2}{2} \|w_{m+1}\|^2, \qquad D_j = \|\partial w_j\|^2 = \tfrac{1}{4} \|\nabla w_j\|^2. \tag{3}$$

The spectral cross terms cancel on summing over *consecutive* indices; a parity-only sum does not have the same endpoint identity. Spatial smoothing at scale ε is defined only on embedded flat disks a controlled distance from the cones. Cone annuli have area $O(\varepsilon^2)$, while cutoff and smoothing derivatives cost $O(\varepsilon^{-1})$. This first proves a uniform-in-mode bound $\|\nabla f_j\| \le C_\lambda \|f\|_\infty$ for a bounded local solution. It does not yet prove summability of those gradients.

The stronger argument selects an orbit segment of length $T = s/\varepsilon$ which avoids a cone neighborhood of radius $L\varepsilon$. Its bad-set measure is $O(s + \varepsilon^2)$. With q_ε the eigenfunction restricted to good initial points, the phase-corrected average is

$$g_\varepsilon = \frac{1}{T}\int_0^T \mathrm{e}^{i\lambda t} q_\varepsilon \circ \Psi_{-t}\,\mathrm{d}t, \qquad \mathcal{A}g_\varepsilon = \frac{q_\varepsilon - \mathrm{e}^{i\lambda T} q_\varepsilon \circ \Psi_{-T}}{T}.$$

This endpoint formula avoids differentiating the indicator of a good set. After permitted smoothing, $\|\mathcal{A}w_\varepsilon\| = O(\varepsilon/s)$ but $\|Y\mathcal{A}w_\varepsilon\| = O(1/s)$. First hold $s > 0$ fixed and let $\varepsilon \to 0$. The latter quantity converges weakly to zero, not by a claimed small norm. Pairing it with a high Fourier tail identifies an identity involving exactly two adjacent endpoint modes. Only after obtaining an exact local solution at fixed s does the source reapply its bounded-mode lemma to recover constants independent of s. Then it takes $s \to 0$. Interchanging these limits would discard the stated control.

The resulting endpoint estimate is

$$\left\|\bar\partial f_m\right\|^2 + \|\partial f_{m-1}\|^2 \le \frac{\lambda^2}{4}(\|f_{m-1}\|^2 + \|f_m\|^2). \tag{4}$$

Summing now gives actual horizontal L^2 derivatives and $\|\nabla f\|^2 \le \lambda^2 \|f\|^2$. Since $Xf = i\lambda f$, orthogonality forces $Yf = 0$. The source does not thereby give a decay rate for the angular coefficient tail or a uniform approximate-eigenfunction estimate.

2.3 Plane waves, reflections, and the spectral endpoint

Inside Q, transverse rigidity implies

$$F(x, v) = c(v)\mathrm{e}^{i\lambda x\cdot v}. \tag{5}$$

For an affine reflection across a side, write $x \mapsto R_j x + d_j$. Comparison in a connected regular crossing chart yields

$$c(v) = \mathrm{e}^{-i\lambda d_j\cdot v} c(R_j v) \quad \text{for almost every } v. \tag{6}$$

This is not an assumed pointwise trace of a measurable initial eigenfunction. Place an irrational-angle vertex at the origin; then $d_1 = d_2 = 0$. The two origin-side reflections generate rotation through 2α. Irrationality of α/π makes c constant by its Fourier coefficients. The third side has $d_3 \neq 0$. For nonzero constant c, $\mathrm{e}^{-i\lambda d_3\cdot v} = 1$ almost everywhere, hence everywhere by continuity, forcing $\lambda = 0$.

For an unbounded L^2 eigenfunction the source uses the injective, phase-equivariant map $F \mapsto F/(1 + |F|)$ to reach the bounded case. It then proves the spectral implication to weak mixing using an invariant product kernel, a nonzero compact Hilbert–Schmidt operator on L^2_0, and a common character on a finite-dimensional invariant eigenspace. This last step is necessary: proving only invariant functions constant would establish ergodicity, not weak mixing. Our finite certificates below concern the reflection endpoint and do not replace any of these global steps.

3 A quantitative reflection obstruction

On $L^2(S^1, \sigma)$, reflection and rotation are unitary. Let the origin-side axes make angle α and let $d = d_3 \neq 0$. For an arbitrary amplitude c, define $c_0 = \int c\,\mathrm{d}\sigma$ and

$$\varepsilon_j = \|c - c\circ R_j\|, \quad j = 1, 2, \qquad \delta = \varepsilon_1 + \varepsilon_2, \tag{7}$$

$$\varepsilon_3 = \left\|c - \mathrm{e}^{-i\lambda d\cdot v} c\circ R_3\right\|, \qquad \eta_K^2 = \sum_{|k|>K} |\widehat{c}_k|^2, \tag{8}$$

$$g_K(\alpha) = 2\min_{1\le k\le K} |\sin(k\alpha)|, \qquad B_K^2 = \delta^2/g_K(\alpha)^2 + \eta_K^2. \tag{9}$$

Here $K \geq 1$ is an integer and $g_K > 0$ is required. Irrationality ensures this for every fixed K, but no uniform positive lower bound as $K \to \infty$. Actual upper bounds for δ, η_K and a lower bound for g_K may replace their exact values in B_K.

Theorem 3.1 (Amplitude defect certificate)**.** *Put $m = \|c\|$ and suppose $B_K < m$. Then*

$$\sqrt{m^2 - B_K^2}\,\beta(|\lambda||d|) \leq \varepsilon_3 + 2B_K, \qquad \beta(t)^2 = 2\big(1 - J_0(t)\big), \tag{10}$$

where $J_0(t) = \int \cos(t\cos\theta)\,\mathrm{d}\sigma(\theta)$ is its integral definition. A strict reverse inequality excludes any amplitude obeying the listed norm, tail, and reflection bounds.

Proof. The product of the first two reflections is a rotation through 2α or its inverse. Unitarity and the triangle inequality give rotational defect at most δ. Parseval therefore yields

$$\sum_{k\neq 0} |1 - \mathrm{e}^{2ik\alpha}|^2 |\widehat{c}_k|^2 \leq \delta^2, \qquad \|c - c_0\|^2 \leq B_K^2.$$

Let $V_\lambda h(v) = \mathrm{e}^{-i\lambda d\cdot v} h(R_3 v)$, also unitary. With $h = c - c_0$,

$$\|(I - V_\lambda)c_0\| \leq \|(I - V_\lambda)c\| + \|(I - V_\lambda)h\| \leq \varepsilon_3 + 2\,\|h\|\,.$$

The left side is $|c_0|\,\big\|1 - \mathrm{e}^{-i\lambda d\cdot v}\big\| = |c_0|\beta(|\lambda||d|)$. Orthogonality gives $|c_0|^2 = m^2 - \|h\|^2 \geq m^2 - B_K^2$. Notice that no reflection invariance of h, and no independence of its phase and orientation, was assumed. □

Lemma 3.2 (An explicit global phase margin)**.** *For every $t \geq 0$,*

$$\sqrt{15/32}\,\min(t, 1) \leq \beta(t) \leq \min(t/\sqrt{2}, 2). \tag{11}$$

For $0 \leq t \leq 2$ the stronger lower bound is $\beta(t)^2 \geq t^2/2 - t^4/32$. Moreover $|\beta(t) - \beta(s)| \leq |t - s|/\sqrt{2}$ for $s, t \geq 0$.

Proof. The inequality $\cos u \leq 1 - u^2/2 + u^4/24$ follows, for example, by integrating four times the nonnegative fourth derivative $1 - \cos u$ of the even difference. The exact moments $\int \cos^2\theta\,\mathrm{d}\sigma = 1/2$ and $\int \cos^4\theta\,\mathrm{d}\sigma = 3/8$ imply the polynomial lower bound. For $t \leq 1$ it is at least $15t^2/32$. On $[1, 2]$ the polynomial $t^2/2 - t^4/32$ is increasing, so it is at least $15/32$. For $t \geq 2$, the law of $x = \cos\theta$ has density $1/(\pi\sqrt{1 - x^2}) \geq 1/\pi$ on $(-1, 1)$. Consequently

$$\beta(t)^2 \geq \frac{2}{\pi}\int_{-1}^{1} (1 - \cos(tx))\,\mathrm{d}x = \frac{4}{\pi}\left(1 - \frac{\sin t}{t}\right) \geq \frac{2}{\pi} > \frac{1}{2} > \frac{15}{32}.$$

The upper bounds and Lipschitz estimate follow from $|\mathrm{e}^{ia} - \mathrm{e}^{ib}| \leq |a - b|$, the same second moment, and the reverse triangle inequality for the two L^2 norms. No numerical Bessel evaluation enters this proof. □

Corollary 3.3 (Frequency floors, not just a sampled band)**.** *Fix $\lambda_0 > 0$. If $m \geq m_*$, $B_K \leq B_* < m_*$ and $\varepsilon_3 \leq e_*$ uniformly over a declared family of candidate amplitudes and frequencies $|\lambda| \geq \lambda_0$, then none exists if*

$$\sqrt{m_*^2 - B_*^2}\,\sqrt{15/32}\,\min(\lambda_0|d|, 1) > e_* + 2B_*. \tag{12}$$

The uniform hypotheses, not the frequency floor alone, do the work here. The bound degenerates as $\lambda_0 \downarrow 0$, as it must in the presence of constant invariant functions. To recover the exact source endpoint, use $\delta = \varepsilon_3 = 0$, let $K \to \infty$ for a fixed c, and use only $\eta_K \to 0$. No quantitative tail rate is required in that exact argument.

4 Angular tails, perturbations, and an obstruction

If c has a certified homogeneous angular Sobolev seminorm $A_r^2 = \sum_{k\neq 0} |k|^{2r}|\widehat{c}_k|^2 < \infty$, $r > 0$, then

$$\eta_K \leq A_r(K+1)^{-r}. \tag{13}$$

This is an additional hypothesis for an approximate certificate; the source's bounded measurable eigenfunction argument does not supply it. Under a separately proved Diophantine bound $|1-\mathrm{e}^{2ik\alpha}| \geq \kappa|k|^{-\tau}$ for all $k \neq 0$, with $\kappa > 0$ and $\tau \geq 0$, there is also the cutoff-free estimate

$$\|c - c_0\| \leq (\delta/\kappa)^{r/(r+\tau)} A_r^{\tau/(r+\tau)}. \tag{14}$$

For $\tau > 0$, apply Hölder to $\sum_{k\neq 0} |\widehat{c}_k|^2$ with factors $(|k|^{-2\tau}|\widehat{c}_k|^2)^{r/(r+\tau)}$ and $(|k|^{2r}|\widehat{c}_k|^2)^{\tau/(r+\tau)}$. The first sum is at most δ^2/κ^2. The case $\tau = 0$ is direct. This is classical interpolation, not a new Diophantine assertion.

Proposition 4.1 (No tail-free irrational coercivity)**.** *For every irrational α/π, there are mean-zero unit amplitudes with both origin-side reflection defects tending to zero. Thus there is no constant $C(\alpha) < \infty$ for which $\|c - c_0\| \leq C(\alpha)(\varepsilon_1 + \varepsilon_2)$ for all c.*

Proof. Take the first side as the horizontal axis. The pigeonhole approximation argument gives integers $q \to \infty$ with $|\sin(q\alpha)| \to 0$. Indeed, partition $[0, 1)$ into N equal intervals and compare the $N + 1$ fractional parts of $j\alpha/\pi$, $0 \leq j \leq N$; irrationality forces an unbounded subsequence of the resulting denominators. Set $c_q(\theta) = \sqrt{2}\cos(q\theta)$. Reflection in the first side fixes it, and direct Fourier calculation gives $\|c_q - c_q(2\alpha - \theta)\| = 2|\sin(q\alpha)|$. It has mean zero and norm one. Its tail is exactly one whenever $q > K$. □

The third-side residual has not been made small by this example. The point is to disprove the missing coercivity step, not to construct billiard eigenfunctions. It also shows why any fixed finite bank of low angular moments misses admissible high-frequency obstructions.

4.1 Explicit geometry and measurement error budgets

If $|\alpha - \widetilde{\alpha}| \leq a$, then

$$g_K(\alpha) \geq g_K(\widetilde{\alpha}) - 2Ka. \tag{15}$$

Use only a strictly positive lower endpoint. If $|d - \widetilde{d}| \leq b$, then $|d| \geq (|\widetilde{d}| - b)_+$ in (12). For an actual $H^1(S^1)$ amplitude, reflection axes differing by a_j give $\left\| c \circ R_j - c \circ \widetilde{R}_j \right\| \leq 2a_j \|\partial_\theta c\|$. On a bounded frequency band $|\lambda| \leq \Lambda$, phase-displacement error adds at most $\Lambda b \|c\|$ to the third residual. This last error is not uniform on an unbounded band. Each statement follows by integrating the angular derivative along the intervening rotations, or by the pointwise Lipschitz bound for the exponential. Merely changing a matrix representing a reflection without controlling the physical operator has no such guarantee.

If a residual is omitted on a measured edge-angle subset of normalized measure ζ, an actual bound $|\operatorname{Tr} F| \leq H$ on both traces gives an additional $2H\sqrt{\zeta}$ in its full norm. Without such a bound, small excluded measure need not mean small L^2 error. Vertex endpoints themselves have zero edge measure; uncontrolled neighborhoods of them do not disappear from a finite certificate by convention.

5 From physical fields to amplitude defects

This section supplies the physical trace estimate absent from a matrix-only certificate. Assume from the start that $F \in H^1(Q; L^2(S^1, \sigma))$ and $\|F\|_{L^2(\mu)} = 1$. No derivative in θ is required here. Put

$$E^2 = \|(X - i\lambda)F\|^2 + \|YF\|^2, \quad u(x,v) = \mathrm{e}^{-i\lambda x\cdot v}F(x,v), \quad c(v) = \mathrm{av}_Q\, u(x,v), \quad w = u - c. \tag{16}$$

Multiplication by this phase preserves the stated spatial Sobolev space for each finite λ. The exact identity

$$\nabla u = \mathrm{e}^{-i\lambda x\cdot v}\big[v(X - i\lambda)F + v^{\perp}YF\big], \qquad \|\nabla u\| = E \tag{17}$$

uses the orthonormal physical directions $v, v^{\perp}$.

Lemma 5.1 (Explicit mean and side estimates)**.** *For each side j let p_j be the opposite vertex and $D_j = \sup_{x\in Q}|x - p_j| \le D$. With $T_j = (D^2 + DD_j)^{1/2} \le \sqrt{2}D$, one has*

$$\|w\| \le DE, \qquad \|\mathrm{Tr}_j\, w\|_j \le T_jE, \qquad \|c\|^2 = 1 - \|w\|^2 \ge 1 - D^2E^2. \tag{18}$$

Proof. We prove the non-sharp but explicit convex-domain Poincaré bound D. For a smooth Hilbert-valued function the variance identity and a segment integral give

$$\|u - \mathrm{av}_Q\, u\|^2 \le \frac{D^2}{2A^2}\int_Q\int_Q\int_0^1 |\nabla u((1-t)x + ty)|^2\,\mathrm{d}t\,\mathrm{d}x\,\mathrm{d}y.$$

By symmetry restrict to $0 \le t \le 1/2$ and multiply by two. For fixed y, the change $z = (1-t)x + ty$ has Jacobian $(1-t)^2$; convexity puts its image in Q. As $\int_0^{1/2}(1-t)^{-2}\,\mathrm{d}t = 1$, the triple integral is at most $2A\int_Q|\nabla u|^2$. This proves the first inequality, also after angular integration. Sobolev density gives it for the stated space.

Let ℓ_j and h_j be the side length and opposite altitude, so $h_j\ell_j = 2A$. The vector field $x - p_j$ has divergence two, normal component h_j on side j, and zero normal component on the other two sides. The divergence theorem gives the exact normalized identity

$$\|\mathrm{Tr}_j\, w\|_j^2 = \|w\|^2 + \mathrm{Re}\,\mathrm{av}_Q\int_{S^1}\overline{w}(x,v)(x - p_j)\cdot\nabla w(x,v)\,\mathrm{d}\sigma(v). \tag{19}$$

Cauchy–Schwarz bounds it by $\|w\|^2 + D_j\,\|w\|\,\|\nabla w\| \le T_j^2E^2$. The identity extends from smooth fields by the Sobolev trace theorem on a triangle and density. Finally c and w are orthogonal in the interior probability norm and $|u| = |F|$, proving the last assertion. □

The actual reflected boundary defect is

$$b_j = \|\mathrm{Tr}_j\, F(x,v) - \mathrm{Tr}_j\, F(x, R_jv)\|_j. \tag{20}$$

It uses the trace of the same field with reflected velocity; it is not a finite list of boundary samples. On that side $R_jx + d_j = x$, and symmetry of R_j gives $x\cdot R_jv = (x - d_j)\cdot v$. Hence

$$u(x,v) - \mathrm{e}^{-i\lambda d_j\cdot v}u(x, R_jv) = \mathrm{e}^{-i\lambda x\cdot v}\big[F(x,v) - F(x,R_jv)\big].$$

Separating $u = c + w$ and using the trace estimate proves

$$\varepsilon_j \le b_j + 2T_jE \quad (j = 1,2,3). \tag{21}$$

Both reflected trace terms have the same norm because reflection preserves σ; no conditional angular uniformity is assumed.

Theorem 5.2 (Physical frequency-exclusion certificate)**.** *Place the vertex between sides one and two at the origin. For a field with the preceding regularity and normalization, take a certified tail bound η_K for the amplitude in (16). Define*

$$\Delta = b_1 + b_2 + 2(T_1 + T_2)E, \qquad B^2 = \Delta^2 / g_K(\alpha)^2 + \eta_K^2.$$

If $D^2E^2 + B^2 < 1$, necessarily

$$\sqrt{1 - D^2E^2 - B^2}\, \beta(|\lambda||d_3|) \le b_3 + 2T_3E + 2B. \tag{22}$$

Replacing β by the lower bound in (11) gives an entirely elementary sufficient test for exclusion when the inequality is strictly reversed. All upper bounds on the right and in the subtracted terms may be enlarged safely while preserving the strict positivity condition.

Proof. Lemma 5.1 gives $m^2 \ge 1 - D^2E^2$; (21) supplies the three amplitude errors. Apply Theorem 3.1, with B an upper bound for B_K. □

The theorem is not a lower bound for $\|(X - i\lambda)F\|$ alone. Its E includes transverse differentiation. It does not assert a spectral gap for a weakly mixing flow, a mixing rate, or the absence of approximate eigenvectors in the Hilbert-space generator domain. The source provides $E = 0$ for exact bounded eigenfunctions only after its singular-set argument. The trace theorem here is a separate quantitative route for already regular candidate fields, not a retroactive assumption in that source proof.

6 An exact irrational-angle triangle

Take

$$Q = \operatorname{int}\operatorname{conv}\{(0,0), (2,0), (2,1)\}. \tag{23}$$

The origin angle is $\alpha = \arctan(1/2)$. It is irrational as a multiple of π: if α/π were rational, $z = \mathrm{e}^{2i\alpha} = (3+4i)/5$ would be a root of unity. Then $z + z^{-1} = 6/5$ would be a rational algebraic integer and therefore an integer, a contradiction. The reflection data are exactly

$$R_1 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad R_2 = \frac{1}{5}\begin{pmatrix} 3 & 4 \\ 4 & -3 \end{pmatrix}, \quad R_3 = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}, \quad d_3 = (4, 0).$$

The first two offsets vanish. Here $A = 1$, $D^2 = 5$ and $T_j \le \sqrt{10} < 4$. For $K = 2$ the squared rotational divisors are

$$|1 - z|^2 = 4/5, \qquad |1 - z^2|^2 = 64/25, \qquad g_2^2 = 4/5.$$

These are rational identities, not floating-point tests of irrationality.

Suppose a normalized H^1 candidate has the *certified* data

$$E \le 1/1000, \qquad b_j \le 1/1000 \ (j = 1, 2, 3), \qquad \eta_2 \le 1/100. \tag{24}$$

Then $\Delta \le 9/500$ and

$$B^2 \le \frac{(9/500)^2}{4/5} + \frac{1}{10000} = \frac{505}{10^6} < \left(\frac{23}{1000}\right)^2.$$

For every $|\lambda| \ge 1/4$, $|\lambda||d_3| \ge 1$. The squared left side of the exclusion inequality is at least

$$\frac{15}{32}\left(1 - \frac{5 + 529}{10^6}\right), \quad \text{whereas the squared right side is at most} \quad \left(\frac{9}{1000} + \frac{46}{1000}\right)^2 = \frac{121}{40000}. \tag{25}$$

The first rational number is strictly larger. Thus there is no such candidate at *any* frequency in that entire range. This is a certificate for the declared residual-and-tail class, not a measured property of every possible eigenfunction. In particular, we did not generate approximate billiard data and pretend it satisfied (24).

For comparison the exact interior plane wave $F = \mathrm{e}^{i\lambda x\cdot v}$ has $E = 0$, $c = 1$, zero origin-side defects, and third-side defect exactly $\beta(4|\lambda|)$. This makes the missing boundary condition visible: solving the interior differential equations is not enough. At $\lambda = 0$ the constant field has all defects zero and is correctly not excluded.

The elementary monomial integrals in this triangle are also exact:

$$\int_Q x^a y^b \,\mathrm{d}x\,\mathrm{d}y = \frac{2^{a+1}}{(b+1)(a+b+2)}, \qquad a, b \in \mathbb{Z}_{\geq 0}. \tag{26}$$

Indeed $0 < x < 2$ and $0 < y < x/2$. Together with normalized affine edge parametrizations and Fourier orthogonality, this gives rational physical Gram entries for the finite constructions below. No area or edge-length factor is hidden in a coefficient norm.

7 Ram moments and a physical residual certificate

7.1 What is transferred from the original corpus

The comparison report TR-item-037 discusses ergodicity and explicitly separates a finite Cheeger or moment certificate from a uniform theorem on the infinite dynamical problem. It is used as a locator and scope comparison, not as a weak-mixing theorem. Fresh original proof passages in TR250 [6], rather than only that comparison, were read for the physical space, moment reconstruction, boundary norm, and interface arguments. They establish three relevant safeguards:

- A moment map is a coordinate map on a specified finite physical space; point jets do not automatically extend to a weak Sobolev space.
- Restricting a dual norm to finitely many tests gives a lower bound, not an upper certificate for an untested physical residual.
- Boundary mismatch needs a trace norm, and a Schur norm with finite inf–sup constant one does not prove uniform stability in physical trace units. Broken gradients alone miss jumps between panels.

The original Schur proof is ordinary Cauchy–Schwarz with a physical mass matrix; its coordinate change is classical, not credited as a new result here. Our additional billiard-specific content is the explicit estimate (19) and its composition with the angular phase obstruction. No new Cheeger bound, roof regularity, or uniform singular-set estimate is extracted from that finite algebra.

7.2 A phase-adapted finite space

For a fixed real λ, choose a *specified* finite space

$$F_a(x,\theta) = \mathrm{e}^{i\lambda x\cdot v(\theta)} u_a(x,\theta), \qquad u_a = \sum_{\ell,k} a_{\ell k}\phi_\ell(x)\mathrm{e}^{ik\theta}, \tag{27}$$

where the ϕ_ℓ are actual $H^1(Q)$ functions and the index set is finite. A conforming polynomial space is one convenient choice. Define

$$G_{\ell k,mh} = 1_{k=h} \int_Q \overline{\phi_\ell} \phi_m \frac{\mathrm{d}x}{A}, \qquad K_{\ell k,mh} = 1_{k=h} \int_Q \overline{\nabla \phi_\ell} \cdot \nabla \phi_m \frac{\mathrm{d}x}{A}.$$

These definitions apply to real or complex spatial bases and give directly

$$\|F_a\|^2 = a^* G a, \qquad E^2 = a^* K a, \qquad \widehat{c}_k = \sum_\ell a_{\ell k} \operatorname{av}_Q \phi_\ell.$$

The exact tail η_K is computed from these mean coefficients, not from all coefficients of u_a. This finite construction has a known largest angular mode; an approximation to an infinite field additionally needs a certified remainder in every norm used.

For a full-column-rank moment map $m = Ba$ with a declared left inverse R on its range, physical squared norms become $m^* R^* G R m$ and $m^* R^* K R m$. Noisy off-range data require a specified reconstruction, rather than an unstated inverse. For example a moment perturbation e has reconstructed L^2 error exactly $(e^* R^* G R e)^{1/2}$ whenever Re is the coefficient error being used. Boundary and angular-tail errors use their own physical operators. These formulas do not convert a finite lower frame bound into an infinite-dimensional bound. A poorly conditioned bank may amplify small moment errors and make (22) inconclusive.

7.3 Certified phase truncation on a side

In the phase-adapted coordinates the exact side defect is

$$b_j = \left\| \operatorname{Tr}_j u_a(x,v) - \mathrm{e}^{-i\lambda d_j \cdot v} \operatorname{Tr}_j u_a(x, R_j v) \right\|_j .$$

The origin sides require no phase approximation. For the third side set $t = |\lambda||d_3|$ and replace the exponential by $p_M(\xi) = \sum_{h=0}^{M} (-i\xi)^h / h!$, evaluated at $\xi = \lambda d_3 \cdot v$. Taylor's integral remainder on the real line gives

$$b_3 \le \|\operatorname{Tr}_3 u_a - p_M(\lambda d_3 \cdot v) \operatorname{Tr}_3 u_a \circ R_3\|_3 + \frac{t^{M+1}}{(M+1)!} \|\operatorname{Tr}_3 u_a\|_3 . \tag{28}$$

The derivative of the exact exponential has modulus one, so no factor e^t is necessary. The remainder is a uniform *operator* bound multiplied by the full trace norm, not by a collection of sampled values. For rational vertices, rational λ, polynomial spatial functions and rational complex coefficients, the fixture's reflection matrices and all truncated integrals can be evaluated exactly by finite Laurent-polynomial arithmetic and (26). Square-root upper endpoints can be checked by rational squaring. The Taylor degree may be expensive when t is large; the global phase lower bound does not remove this cost of measuring a candidate's actual residual.

More generally an enclosure for a squared norm matrix must be converted to a quadratic-form upper bound. If a Hermitian computed matrix $\widetilde{H}$ satisfies $|H_{ij} - \widetilde{H}_{ij}| \le e_{ij} = e_{ji}$, then

$$a^* H a \le a^* \widetilde{H} a + \sum_{ij} e_{ij} |a_i||a_j| \le a^* \widetilde{H} a + \sum_i \left(\sum_j e_{ij} \right) |a_i|^2.$$

The last inequality is $2|a_i||a_j| \le |a_i|^2 + |a_j|^2$. Use the actual coefficient vector or a justified bound for it. A small entrywise error without its dimension and coefficient cost is not a small residual certificate. If a trial field is assembled from nonconforming panels, reconstruct a conforming field and bound its error, including trace jumps; zero panelwise derivatives of opposite constants cannot pass as a small physical Sobolev defect.

8 Extensions and exact limitations

8.1 Uniform families and frequency coverage

On a family of nondegenerate triangles the constants can be bounded using upper diameter bounds, lower $|d_3|$, and positive certified finite divisors g_K. Approaching a rational angle with denominator at most K destroys that divisor bound. Degenerating geometry may also destroy useful margins. Each field must use its own normalized area and edge measures. The source does not assert that its cone-smoothing constants are uniform under such a degeneration, and our proof does not need that assertion.

An interval in frequency can instead be covered by rigorous cells, using the Lipschitz bound for β and explicit bounds for changes in the candidate defects. Merely evaluating endpoints or a grid is insufficient without those bounds. Corollary 3.3 avoids a phase-grid altogether but still requires uniform control of the other quantities. A sequence of finite cutoffs with no tail estimate proves neither all-frequency exclusion nor weak mixing.

8.2 Return maps and suspension roofs

The report works with the flow identity (1); it does not replace it by an unweighted return-map invariance equation. On a transversal with return map T and roof τ, an eigenfunction restricts, where the return is defined, to $h(Tx) = \mathrm{e}^{i\lambda\tau(x)}h(x)$ with consistent forward-time convention. A proof through that representation must address the singular set, integrability and domain of the roof, and the exact eigenphase. None of those obligations follows from a finite transfer matrix. Our reflection defects keep their physical phase $\mathrm{e}^{-i\lambda d\cdot v}$ for exactly this reason; dropping it would lose the nonzero-frequency obstruction.

8.3 Advantages, costs, and research directions

The main advantage is a named-region, named-norm certificate with every loss explicit: interior defect DE, side defect $2T_jE$, low-mode divisor g_K^{-1}, angular tail η_K, and a global third-side phase margin. It requires no numerical Bessel oracle and has an exact rational geometric example. Its limitations are equally material. The mean-value and trace constants are deliberately elementary rather than optimal. A weak angular tail or a small divisor can overwhelm the margin. Measuring large-frequency boundary phases can be costly even though the phase lower bound stays positive. Most importantly, the transverse residual is independent data.

Useful future work is to obtain a quantitative analogue of the source's ordered-limit transverse rigidity under stated extra regularity; optimize the triangle Poincaré and side constants with certified upper bounds; and build adaptive moment banks that control the *physical* angular tail instead of just retained coefficients. These are research tasks, not outputs of the current finite tests. No rate of decay of correlations, quantitative mixing, or universal Diophantine estimate is claimed.

9 Public status and evidence boundaries

9.1 Current primary literature, October 8, 2026

Chaika–Forni's published theorem concerns a G_δ dense set of non-rational polygons and full Liouville weak mixing, not every irrational triangle. The Annals publisher records online publication on May 1, 2026, volume 203, pages 695–735 [2]. Arana-Herrera–Chaika–Forni study weak mixing in almost every direction for rational billiards; their quantifiers are different [3]. Forni–Moll's primary

abstract distinguishes ergodicity of the holonomy foliation from horizontal cohomological results under simultaneous Diophantine assumptions [4]. None of these abstract statements is used here as a substitute for the source's transverse argument. The private comparison TR037 is not an independent public result proving the present source claim. Earlier catalogue items about ergodicity likewise do not settle its nonzero eigenfrequencies.

The public refresh read publisher/arXiv abstracts and metadata, not all external proofs. The supplied October 5 manuscript makes the stronger all-irrational-triangle claim. This report analyzes that supplied proof and proves the stated local refinements; it does not label the claim a separately established consensus result on the strength of a literature abstract.

9.2 Literal reading and reproducibility

The eight public text files total 1,074 lines: README 18, main 43, bibliography 118, and five scientific modules totaling 895. Their entire literal contents were read in bounded displays. The private comparison TR037 was read only at lines 771–900 and 1300–1468. The freshly read original TR250 passages are lines 121–181 and 564–726; they include the actual norm and Schur proofs and trace qualifications. The remaining lines of those private files and the rest of the corpus are explicitly unread. Heading searches are locators, not proof reads. The owned reading ledger records literal intervals, unread complements, and before/after hashes.

The independent author script checks rational complex Fourier identities, low-mode/tail inequalities, the irrational-angle fixture's reflection data, exact polynomial mean/trace identities, finite side-phase remainder budgets, and the strict rational exclusion margin. The author run passed 1,658 finite cases containing 9,013 exact assertions; its receipt states each case unit. These finite tests supplement the proofs; they do not test an infinite set of frequencies, certify irrationality by sampling, reproduce the cone-limit argument, or establish weak mixing. Independent cross-review counts are not added to the author count. Source snapshot authentication is provenance, not mathematical verification.

10 Conclusion and contribution assessment

The principal local result is Theorem 5.2: a fully normalized physical defect-to-frequency obstruction whose angular, geometric, and boundary hypotheses can be checked separately. The global elementary phase margin and exact rational example make it stronger than a qualitative restatement of the final reflection contradiction. The small-divisor counterexample and the transverse-residual gate delimit exactly what it does not establish. Poincaré estimates, trace identities, Fourier orthogonality, and moment-coordinate transfer are classical ingredients; no historical novelty for those tools is asserted.

The demonstrated increment supports C3 as a notable local quantitative contribution, independently of applicability L3. It is not a new global billiard theorem. The full singular-set domain argument, the ordered limits producing $Yf = 0$, and the all-frequency spectral conclusion remain the source obligations. Neither finite rank nor accurate sample values replace them.

References

[1] *Weak mixing of triangular billiards with an irrational angle.* Supplied public manuscript, October 5, 2026. Cached build: `main.tex`; `introduction.tex`, `geometry.tex`, `energy.tex`, `rigidity.tex`, `conclusion.tex`.

[2] J. Chaika and G. Forni, *Weakly mixing polygonal billiards*, Annals of Mathematics **203** (2026), 695–735. Published online May 1, 2026. https://annals.math.princeton.edu/2026/203-3/p01.

[3] F. Arana-Herrera, J. Chaika and G. Forni, *Weak mixing in rational billiards*, arXiv:2410.11117, 2024. https://arxiv.org/abs/2410.11117.

[4] G. Forni and N. Moll, *Cohomological equation for geodesic flows on flat surfaces*, arXiv:2510.18128, 2025. https://arxiv.org/abs/2510.18128.

[5] M. Subbarao, *Ergodicity and the Scope of Ram Methods*, private comparison report TR-item-037-astra. Selected passages only; confidential corpus consulted locally.

[6] M. Subbarao, TR250, original compatible variational Ram-panel report, revision-one file `TR-rev-1-old-tr250.tex`. Selected original proof passages on physical norms, source/trace mismatch, Schur stability, and trace repair; confidential corpus consulted locally.